

Virtual Reality Game-Based Classification of Arachnophobia: A Two-Step Clustering Approach

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Abstract—Fear is a multifaceted emotion, challenging to define and assess accurately. Biologically, it is an innate response to threats, whereas psychologically, it is shaped by individual experiences and societal influences, sometimes evolving into specific phobias. One such phobia, arachnophobia (the fear of spiders), is particularly widespread and can vary widely in intensity among individuals. This paper presents a prototypal system that classifies the severity of arachnophobia using Machine Learning (ML) algorithms within a Virtual Reality (VR) game-based environment. The proposed system utilizes a two-stage clustering approach to analyze the behavioral data collected during VR exposure. Additionally, participants' self-reported fear levels are measured using the Spider Phobia Questionnaire (SPQ), to provide a comprehensive assessment. Preliminary results suggest that this method could be effectively used to classify arachnophobia intensity level, thus offering potential applications in both clinical settings and video games.

Index Terms—Arachnophobia, Virtual Reality, Clustering, Behavioral Approach Test

I. INTRODUCTION

Fear is a complex emotion subject to various interpretations across different times and fields [1]. Its study is deeply rooted in emotional theory and is influenced by both biological rules and psychological conditioning [2]–[4]. Biologically, fear acts as an evolutionary safeguard, a response that developed over millennia to alert organisms to danger and ensure their survival. Psychologically, fear is shaped by personal history and social context, resulting in a nuanced interplay between instinct and cognition, influenced by genetics, upbringing, and cultural background [1], [4].

When fear becomes irrational and overwhelming, it can develop into a phobia, an exaggerated pathological fear response. Phobias are specific, involving intense and persistent fear of particular objects, situations, or organisms. Despite understanding that these fears are irrational, individuals with phobias

engage in avoidance behaviors, perceiving threats where none exist [5], [6]. The *Diagnostic and Statistical Manual of Mental Disorders* (DSM) provides guidelines for diagnosing phobias, emphasizing marked fear or anxiety in response to specific stimuli [6]. Understanding the mechanisms underlying fear and phobia is crucial for advancing the knowledge of human psychology and developing effective therapeutic interventions.

According to the DSM, specific phobias are categorized into animal, environmental, blood injection injury, and situational types. An additional category, “other,” is used for phobias that do not fit the aforementioned groups. This categorization helps to group phobias with similar behavioral patterns. Arachnophobia, an intense fear of spiders, falls under the category of animal phobia. A study of 7,076 Dutch citizens found that 12.6% experienced animal fear and 26.5% of this group exhibited phobic forms (3.3% of the total) [7].

Arachnophobia often triggers a fight-or-flight response, which may manifest as a “fight” reaction, characterized by aggressive actions to counter the perceived threat, or as a “flight” reaction, involving attempts to evade or distance oneself from danger. Notably, pronounced avoidance behaviors are typically observed only in individuals with severe forms of phobia [8]. Arachnophobia is also associated with disgust [9], [10], with avoidance being a common reaction. However, studies such as [11] suggest that, while spiders can provoke disgust, this reaction is not necessarily tied to phobic fear. Thus, while spiders may elicit disgust-related responses, it is not possible to definitively classify arachnophobic individuals based solely on their levels of disgust.

The objective of this study was to develop a system to classify arachnophobia levels using ML algorithms in a VR environment. The focus was on exploring the feasibility of accurately classifying arachnophobia levels using VR-related data, which allows for realistic and immersive behavioral tests. This method reduces the risk of exposing arachnophobic individuals to actual phobic elements, while providing valuable

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data for assessment. The system incorporates data from behavioral movement tracking, eye tracking, and facial expression analysis, thereby enhancing the classification process with motion and behavioral insights.

Beyond its relevance for clinical research, a system capable of classifying arachnophobia levels can also inform the design of VR games. Adapting in-game experiences based on players' fear responses could enhance immersion and engagement, particularly in genres like horror or survival. Spiders were chosen as a case study due to the high prevalence of arachnophobic subjects and its suitability for realistic simulation in virtual environments, making it an effective model for studying fear in a controlled setting and the efficacy of our system.

The remainder of this paper unfolds as follows: Section II provides an overview of the current state of the art and background information. Section III outlines the methodology and development process of the behavioral test. In Section IV, the test and data collection procedures are described. The analysis of the data is presented in Section V. Section VII presents the preliminary findings. Finally, Section VIII discusses the conclusions drawn from this study.

II. STATE OF THE ART

The assessment of arachnophobia has evolved significantly with advancements in both psychological assessment tools and therapeutic interventions. Behavioral Approach Tests (BAT) are widely recognized as a cornerstone in the assessment of arachnophobia [11]–[13]. These tests are meticulously designed to systematically expose participants to spider stimuli within controlled environments. The aim is to elicit and observe behavioral reactions, particularly avoidance behaviors.

In recent years, the advent of advanced sensor technologies has revolutionized assessment approaches in arachnophobia research [12]. By integrating sensors into BAT setups, researchers can capture a broader spectrum of participant reactions, ranging from physiological responses to subtle behavioral cues. Moreover, researchers leverage facial expression analysis and gaze tracking methods to delve deeper into the intricacies of arachnophobia [10], [11], [14]. By analyzing participants' facial expressions and gaze patterns during spider exposure, it is possible to gain insights into the physiological and behavioral reactions associated with arachnophobia. These non-invasive techniques serve as invaluable tools in unraveling the underlying mechanisms of this phobia, offering a more comprehensive understanding of its manifestations.

Although exposure-based treatments are among the most widely used for their effectiveness in treating phobias, they have several limitations. Firstly, many patients are reluctant to expose themselves to the phobic stimulus. Additionally, *in vivo* exposure cannot be fully controlled by the therapist. Finally, this approach often requires therapists to guide patients, resulting in significant time investment for both parties.

Virtual Reality Exposure Therapy (VRET) has emerged as a promising solution to these challenges. In psychotherapeutic contexts, VR can be described as an experiential form that is as effective as reality in inducing emotional responses [15].

VRET is promising in the treatment of various phobias, including arachnophobia [16]. Powers et al. [17] have demonstrated that VRET can be more effective than traditional *in vivo* treatments. This therapeutic approach allows individuals to confront their fears gradually, starting from less intense stimuli and progressing to more challenging ones. VRET provides a safe and controlled setting for individuals to confront and overcome their phobias, without the risks associated with real-world exposure. One of the key advantages of VRET is its ability to tailor exposure scenarios to each individual's specific needs and fears. Therapists can design virtual environments that closely mirror real-world situations known to trigger phobic reactions, enabling highly targeted and effective treatment. Furthermore, VRET can be administered remotely, which reduces the necessity for therapists to be physically present during sessions. This not only saves time and resources but also enhances accessibility for patients who might struggle with attending in-person therapy. The realism and immersion offered by VRET further amplify its effectiveness. The heightened sense of presence elicits stronger emotional responses, facilitating the therapeutic process and promoting habituation to phobic stimuli.

The proposed system aims to build on these advancements by automating the classification of arachnophobia levels using data from both virtual environments and headset sensors. The integration of traditional and advanced assessment methods seeks to refine arachnophobia evaluation and foster more effective treatment strategies for this widespread phobia. Moreover, the ability to automatically classify arachnophobia levels could inspire innovative video game mechanics that dynamically adjust fear-inducing elements in real-time, thereby enhancing players' engagement.

III. METHODOLOGY

We selected the Meta Quest Pro¹ to provide an immersive environment due to its array of sensors and services. Unlike other devices produced by Reality Labs, Quest Pro incorporates additional sensors to track facial expressions and gaze. While primarily designed to enhance virtual avatar realism, this functionality can also be used to gather behavioral data. Facial expressions are classified on the basis of a set of weights correlated with a set of action units, commonly used in literature for facial tracking [18]. Gaze tracking, on the other hand, is provided as the position and orientation of each eye, based on the selected reference system.

The behavioral test on arachnophobia was conducted in an *ad hoc* VR environment. While physical settings offer precision, they come with complexities and the risks associated with the handling of living spiders. In contrast, VR environments provide a simpler and more extensive data collection, at no risk. By ensuring immersion and maintaining illusions of presence and plausibility [19], results comparable to those of real-world experiments can be achieved.

¹<https://www.meta.com/en/quest/quest-pro/>

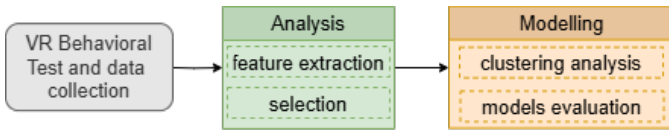


Fig. 1. Representation of the different steps of the reported study. Data were collected during the VR behavioral test with users. The data were then analyzed and processed in order to be used for the clustering classification.

ML was then employed to automatically classify arachnophobic individuals using data collected from the VR behavioral test. Without prior self-assessment knowledge or participant labeling, unsupervised learning emerged as the ideal choice, eliminating the need for pre-existing data labels. Clustering algorithms facilitate the grouping of participants based on their observed reactions during the VR test to identify patterns or relationships without prior labels [20]. An overview of the proposed system is presented in Figure 1.

A. VR Behavioral Test Design

To provide an unbiased classification of arachnophobic subjects, it is crucial to minimize irrelevant phobic elements (e.g., an abandoned cave could induce fear independently from the presence of spiders) in the virtual environment while simultaneously preserving plausibility. Environmental credibility was paramount; thus, only places where one could meet spiders in actual life were included in the virtual environment. In addition, a detailed simulation of spider behavior and interactions added authenticity. The selection of a contemporary house as the main setting provided a plausible and “neutral” backdrop for spider encounters, free of undesired fear-inducing elements.

1) *Phobic Stimuli*: The selected phobic stimuli were the spiders and spiderwebs. To link the fear of spiders with their speed and unpredictability, various types of behavior were simulated and assigned to different spider categories:

- *Escape*: This type of spider waits for the tester to reach a specific distance before fleeing in a predetermined direction, typically hiding behind furniture or crawling on a wall. The possibility that the spider had actually leaped onto the participant was excluded.
- *Patrol*: Patrol spiders move cyclically across the environment, following a predetermined path, but without interacting with the tester.
- *Dangle*: These spiders can descend and retreat from the ceiling using a spiderweb. When the tester reaches appropriate positions in the environment, several spiders descend from the ceiling, while others retract towards it, creating diverse but convincing behaviors.
- *Idle*: Some spiders in the scene simply remain stationary at specific locations.

All types of spider feature credible animations for each assigned behavior. Spiderwebs were developed to be fully interactive, allowing participants to interact with them through the physics of the virtual world. Once the subject touches a spiderweb filament, it breaks.

2) *Mapping of test phases on the spaces in the virtual environment*: To ensure the proper execution of the behavioral test, a set of phases was defined, along with how the subject can progress from one phase to the next. Phases should provide gradually increasing exposure to the phobic stimulus, allowing for the collection of information on the subject’s behavior in increasingly intense situations. The interior of a house provides a credible and effective scenario to split the experiment into phases, as shown in Figure 2, by using a series of locked rooms. In addition to the semantic division into distinct phases, the rooms isolate the phobic stimuli within them, preventing situations created in one place from significantly influencing other sections of the experiment. These physical constraints enable the gradual presentation of the stimulus to the subject, preventing an unintentionally premature exposure to subsequent phases of the experiment.

The order of phase execution and room exploration is regulated by the use of a set of keys. Closed rooms can be opened using a key of the same color, “strategically hidden” (i.e., near a spider) inside the previous room. The use of keys for the transition between phases encourages the exploration of species, thus increasing the overall effectiveness of the experiment. To ensure the comparability of the test across different participants, the keys are systematically hidden in the same positions for each experiment.

In the Starting Phase (Figure 2(a)), participants are introduced to the virtual environment without encountering phobic stimuli. This phase is designed to help testers, especially those unfamiliar with VR technology, to adjust to the device, while exploring the house foyer, kitchen, and living room, all free from spiders. The Second Phase (Figure 2(b)) introduces mild phobic stimuli, in the form of spiderwebs. Participants needed to retrieve a key hidden among the spiderwebs, allowing researchers to assess their initial reactions to subtle exposure to phobic elements. In the Third Phase (Figure 2(c)), the participants encountered spiders for the first time in a controlled setting. Located in a bedroom, the spiders were positioned to avoid direct contact with the tester. Participants must carefully navigate the room to retrieve a key placed under the bed and thus surrounded by spiderwebs. The Fourth Phase (Figure 2(d)) intensifies spider exposure by placing them along the path to the key. In addition, some spiders exhibit dynamic behaviors, such as descending from the ceiling or leaping, to elicit stronger reactions from participants. Finally, in the Fifth Phase (Figure 2(e)), set in the attic, testers face the highest level of spider exposure. A group of crawling spiders is active as participants enter the attic and testers must retrieve the key from atop a crate surrounded by spiders. Participants were required to navigate through this dense concentration of spiders to complete the experiment. Notably, some spiders jump near the participant or drop from the ceiling nearby, hence increasing the exposure level and the tension.

IV. DATA COLLECTION

Data were collected at 50Hz sampling frequency and involved various elements:



Fig. 2. In panel 2(a), the house foyer, as the first phase of the experience, is shown. In panel 2(b), the second phase environment, the office, is illustrated. Panel 2(c), represents the third phase of the experiment. In panels 2(d) and 2(e), the closet and the attic are shown, representing the fourth and the last phases respectively.

- *Face*: the HDM produces 63 weights aligned with the Facial Action Coding System (FACS) [21]. Each weight corresponds to the deviation of the facial area from the neutral position and is normalized between 0 and 1.
- *Eyes*: Gaze is tracked across three reference systems: head, headset, and world, respectively accounting for eye movements, head movements within the virtual environment, and real-world motion.
- *Motion*: The tracking system captures data related to head and hands real-world movements, including position, rotation, velocity, and acceleration.
- *Interaction*: This component monitors physical user interactions using controllers, such as button presses, touches, or proximity, with floating-point values indicating the level of pressure applied.
- *Room*: The system logs the rooms visited within the virtual environment, providing an appropriate context for participant behavior.
- *Entity*: Spider and spiderweb positions are tracked to determine the distance from the phobic stimulus.

A. Experimental Procedure

Data collection was conducted in a controlled laboratory setting. Prior to the VR behavioral test participants signed a data processing consent form because of the sensitive nature of the study, which explicitly stated that the experiment was centered on arachnophobia and would involve exposure to virtual spiders. The recruitment process aimed at diversity by including individuals with varying degrees of arachnophobia. The participants were reminded that they could withdraw from the experiment at any time if they felt uncomfortable. A total of 21 participants were recruited, of whom only one was female. The majority (17 or 78.4%) was between 18 and 25 years old. Approximately half of the participants (9 or 42.9%)

expressed an interest in the horror genre, while most had little to no prior experience with VR.

In the next phase, the participants were briefed on how to navigate and interact within the virtual environment. Their task was to locate various keys scattered throughout a virtual house, unlock different doors, and ultimately exit the building. To collect authentic reactions, testers were not informed of the incremental exposure to spiders they would have experienced in the different phases of the test. None of the participants reported experiencing motion sickness [22], either during or after the VR session.

After the VR test, participants completed a demographic questionnaire followed by the SPQ [23]. The duration of the overall experience varied depending on the participant's familiarity with VR technology and their level of arachnophobia. On average, the behavioral test alone lasted approximately 6 minutes, with some instances extending beyond 14 minutes. The total duration of the experiment, including the behavioral test, questionnaires, and preparation phase, did not exceed 30 minutes. No monetary compensation was provided.

B. Questionnaires

After completing the behavioral test, participants were asked to complete two questionnaires:

- 1) *Demographic questionnaire*: demographic information was gathered such as age, gender, and experience with VR gaming. The data collected helped categorizing the participants and identifying potential factors that influence their levels of arachnophobia.
- 2) *SPQ*: It has been used to assess participants' arachnophobia levels through self-assessment. SPQ consists of 31 statements to which participants respond as true or false. Each response indicating fear is assigned a score of one, whereas responses indicating no fear receive a

score of zero. The total score ranges from 0 to 31, with higher values reflecting greater fear.

Table I presents the results obtained from the SPQ. This evaluation provides a reference point to validate the classification obtained through clustering, but it is not used to label the collected data.

V. DATA ANALYSIS

Data was pre-processed before further analysis. Missing or corrupted data were deleted through interpolation, and categorical information was converted into numerical values. The data were then segmented according to the phases of the behavioral test, ensuring that similar contexts were grouped together for a more accurate analysis.

Subsequently, dimensionality reduction techniques were applied to reduce the dataset. The refined set of features was then used to feed a model, which generated labels indicating each individual's group membership. Finally, the model's performance was validated by comparing its output with self-assessments gathered from the questionnaire.

A. Time Series as Features

Clustering algorithms was directly applied to groups of raw data time series. To speed-up the clustering procedure, the length of each time series was reduced by using two methods tailored for this purpose.

The first method, called *Piecewise Aggregate Approximation* (PAA) [24], involves subsampling a time series by dividing it into an arbitrary number of segments, each with a fixed length. Subsequently, the mean is calculated for each segment, effectively downsampling the time series. The second method, *Symbolic Aggregate approXimation* (SAX) [25], was applied after PAA and quantizes into symbols the means previously obtained. During quantization, symbols are calculated to be equiprobable, based on the assumption of a standard normal distribution.

B. Feature Selection

Feature selection was then performed to reduce the number of features provided to the model, while retaining as much relevant information as possible. Due to the entirely unsupervised nature of the process, the choice of feature selection methods was constrained, precluding guidance based on prior output information.

The first feature selection method involved removing information with variance between participants below a certain threshold. To achieve this, features of all participants were concatenated, considering both the data type and the categorized phase. Subsequently, the variance of each group was calculated, and information below the selected threshold was discarded, thereby reducing the total number of features.

Principal Component Analysis (PCA) [26], a technique for dimensionality reduction, was then used to select a predefined number of features for each phase. The division into phases was preserved during feature selection to tailor the selected feature group for each phase. Following feature selection, clustering was performed to potentially characterize participants.

VI. MODELS

The clustering algorithms used in this study can be separated into two categories based on the accepted input format: *single-point clustering* and *time series clustering*. Table II lists algorithms belonging to the time series clustering category, while Table III lists algorithms belonging to the single-point clustering category.

A. Two-Step Clustering

In this project, since the aim is to group subjects using information from different phases, adopting a single standard algorithm is not feasible. Context separation should facilitate the grouping of similar subjects. Consequently, a *two-step clustering approach* was employed, allowing the use of standard algorithms with slight modifications. In detail:

- *Phase-wise voting*: In the first step, each clustering algorithm from Table II is tested in turn. The algorithm is independently applied to each phase of the behavioral test to group only information from the same context. At the end of the first step, a possible grouping of individuals is proposed for each phase, resulting in a vote among the phases for the assignment of each individual's group.
- *Vote agglomeration*: The objective of the second step is to gather the votes from each phase of the test to propose a final judgment on the group membership of each individual. The final clustering is performed considering the votes generated from the previous step as characteristic features of each individual. The models used for single-point clustering are reported in Table III. *Defined clusters* refer to algorithms where the number of clusters is predetermined or known, while *undefined clusters* involve algorithms where the number of clusters is not predefined and is determined based on the data itself. This enables the application of a clustering algorithm from the single-point clustering category, which takes individual data points as input, to obtain the final cluster. At the end of vote agglomeration, several groups should form, based on the number of clusters set as a parameter. The groups should contain subjects with similar levels of arachnophobia. For example, with two clusters set as a parameter, there will be a clear division into two groups between arachnophobic and nonphobic individuals.

This approach ensures that clustering accounts for variations across different phases, while ultimately grouping individuals based on their overall behavioral patterns.

B. Model Validation

Since two-step clustering is an unsupervised process, there are no labels for model training and validation. The clustering results, which estimate group membership, can be compared to self-assessed levels of arachnophobia, gathered through the SPQ. This assessment serves as a reference, considering the difficulty in objectively determining each individual's phobia level. Given the challenges of identifying a clear pattern or setting an arbitrary threshold to reliably categorize individuals based on self-assessments, a *single-point clustering model* was

TABLE I
RESULTS OF THE SPQ FOR EACH SUBJECT

subject	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
SPQ	2	5	5	5	16	1	16	5	20	4	10	11	2	10	5	1	7	10	12	11	5

TABLE II
ALGORITHMS BELONGING TO THE TIME SERIES CLUSTERING CATEGORY

Algorithm Type	Subtypes
Time Series KMean	Euclidean, DTW
Kernel KMean	None
KShape	None
Hidden Markov Model	Gaussian, Gaussian Mixture Model

TABLE III
ALGORITHMS BELONGING TO THE SINGLE-POINT CLUSTERING CATEGORY

Cluster	Algorithm Type
Defined Clusters	Agglomerative
	KMean
	Birch
	Bisecting KMean
	Spectral Clustering
Undefined Clusters	Gaussian Mixture Clustering
	Optics
	MeanShift
	Affinity Propagation

chosen. Similar to the data obtained from the simulation, grouping can be performed using self-assessments as the input features. This approach avoids the risk of incorrect validation that could arise from selecting an inappropriate threshold to divide participants. By using clustering, it is thus possible to group self-assessments in such a way that they are plausibly comparable with the results of the two-step clustering. Single-point clustering with SPQ scores has been employed to form the final groups. This method is suitable for our application because it allows the formation of natural groupings based on the characteristics encoded in voting data.

After determining potential real groupings from the questionnaires, the model validation can proceed by calculating the similarity between these groupings and those obtained from the simulation. This is done using the *Adjusted Rand Index* (ARI), which measures the similarity between two clusters. The ARI calculation begins with the *Rand Index* (RI), which assesses the proportion of element pairs that are consistently clustered in both groupings:

$$RI = \frac{a + b}{\binom{N}{2}} \quad (1)$$

where a is the number of pairs in the same cluster in both groupings, b is the number of pairs in different clusters, and N is the total number of elements. The RI ranges from 0, indicating that no pairs are consistently classified, to 1, representing identical clustering. To adjust for chance, the *Expected Rand Index* (ERI) is calculated:

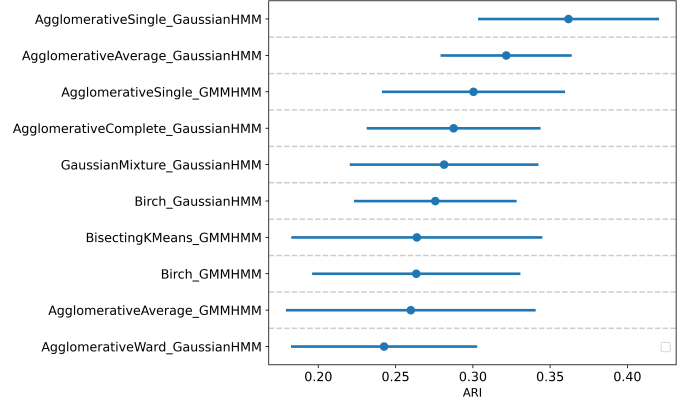


Fig. 3. Performance comparison of different models using SA clustering with the SPQ instrument. Results of the top 10 pairs of clustering models with two clusters. Error bars represent standard errors.

$$ERI = \frac{\sum n_i^2 \sum n_j^2}{N^2} \quad (2)$$

Finally, the ARI is derived using:

$$ARI = \frac{RI - ERI}{1 - ERI} \quad (3)$$

The ARI ranges from -1 to 1, where values close to 1 indicate a strong match between clusters, values near 0 suggest random clustering, and values near -1 indicate complete disagreement.

VII. RESULTS

The results obtained from applying SA clustering with the SPQ tool are depicted in Figure 3. This dominance of the agglomerative category in classification tasks could be attributed to the inherent structure of the two-step clustering method. The second phase of this approach entails aggregating votes obtained from the initial step, a process well-suited to the SA clustering paradigm. This dominance underscores the importance of selecting appropriate clustering algorithms for effective two-step clustering.

A. Parameters Analysis

To assess factors influencing performance, analysis was conducted focusing on key parameters. As shown in Figure 4, increasing the number of clusters notably diminishes average performances, particularly for the most effective pairs. However, for less advantageous pairs, dividing subjects into three clusters may be preferable, indicating that a limited number of clusters, containing similar levels of arachnophobia, can be extracted from both data and questionnaires. The best grouping

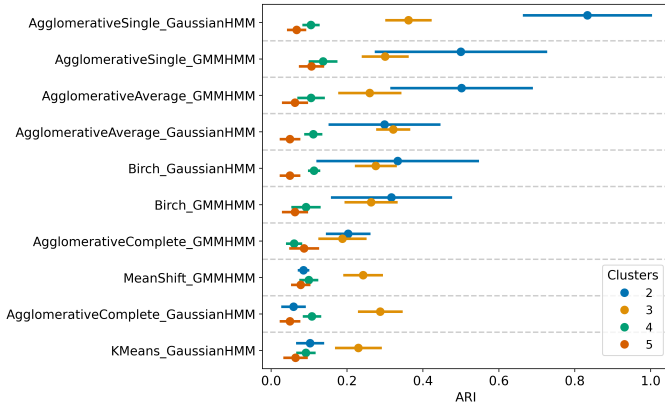


Fig. 4. Focus on SA model for SPQ, and different number of clusters. Results of the top 10 pairs of clustering models. Error bars represent standard errors.

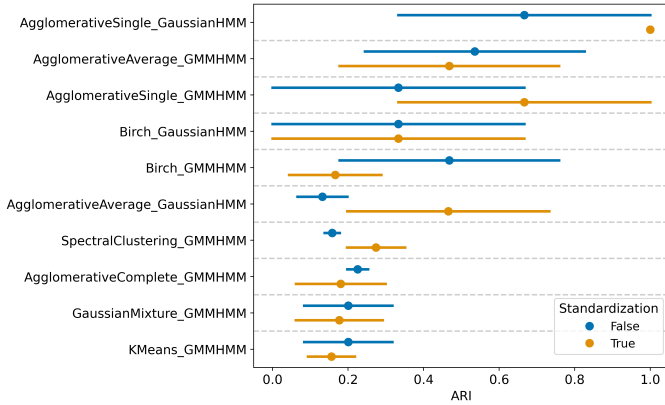


Fig. 5. Focus on SA model for SPQ, performance comparison with and without standardization. Results of the top 10 pairs of clustering models. Error bars represent standard errors.

was achieved when attempting to divide the subjects into two distinct groups, separating arachnophobic subjects from non-phobic subjects.

An investigation was conducted to determine if meaningful results could be obtained from time series by standardizing before selection. This involved dividing the individuals into two clusters and validating the AS model with the SPQ. Figure 5 demonstrates that, on average, standardization before the feature selection positively influences model performances. This could be attributed to the use of PCA, which benefits from data standardization.

Figure 6 illustrates the application of optimal parameters for each number of features, examining both the performance in a favorable scenario and the performance change based on the number of selected features. Notably, pairs containing the SA model consistently exhibited high performance, with the Gaussian HMM model achieving clustering identical to the ideal scenario, irrespective of the number of features considered. Other top-ranking pairs also yielded excellent results, and an adequate number of features were selected. Analysis of the cluster structures indicated that the algorithm tended to isolate individuals with the highest recorded levels

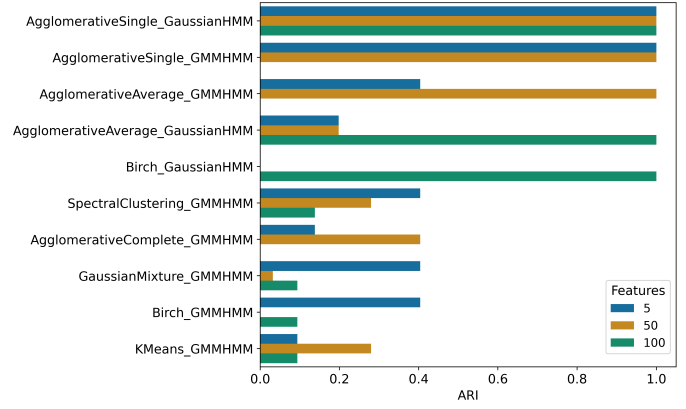


Fig. 6. Performance comparison with optimal parameters using SA clustering with the SPQ tool, varying the number of selected features.

of arachnophobia, as reflected in the SPQ questionnaire scores.

The analyses indicate that the selected parameters maintain the superiority of the agglomerative category for most values, without significantly altering the ranking of the best algorithms for classification.

VIII. DISCUSSION AND CONCLUSIONS

This project laid the groundwork for developing a system to automatically and safely spot individuals with arachnophobia by integrating VR with unsupervised learning techniques. The goal was to create a method that could identify arachnophobic individuals in a controlled virtual environment, thereby eliminating the need for self-assessment tools and real-world exposure. The results suggest that the design of the virtual environment plays a crucial role in the system's performance. Crafting a VR simulation that adheres to specific guidelines was challenging but essential for eliciting genuine reactions from participants. The virtual environment was carefully designed to balance complexity and realism while avoiding simple or intricate settings. A house setting was chosen to minimize irrelevant phobic stimuli, ensuring that reactions could be reliably attributed to arachnophobia. Additionally, interactive elements, such as spiderwebs that break upon touch, contributed to the authenticity of the experience. BAT effectively served as an intermediary, linking data collection to the model. Phased exposure to stimuli helped contextualize the data, and the use of isolated rooms connected through a house layout prevented premature termination of the experiment due to highly phobic stimuli.

Although this study focused on arachnophobia, this approach lays the foundation for broader applications. VR and unsupervised learning could be adapted to classify and analyze various phobias, with potential uses in both clinical and entertainment contexts. In therapeutic settings, similar VR-based methodologies could aid in diagnosing and treating specific phobias, offering controlled exposure without real-world risks. In entertainment, the system could create adaptive experiences in video games, dynamically adjusting challenges based on a user's fear response.

Two-step clustering was employed to aggregate the data from different phases. The agglomerative model with single linkage criteria performed particularly well, probably because of its suitability for data agglomeration. This model was also used to validate the clusters obtained from self-assessments, despite the unsupervised nature of the process. The similarity between clusters and self-assessment results was assessed using ARI, highlighting the need for clusters to originate from similar models for effective comparison. The results show that using preprocessed time-series data can significantly improve performance compared to subsampling methods. This underscores the importance of analyzing and comparing complete datasets. However, this also indicates that the model's applicability may be limited to scenarios created under strict conditions, as shown in this study.

Future work should involve expanding the dataset to include larger and more diverse groups of participants. This would help to generalize the findings and validate the hypotheses, especially regarding the extraction and classification of individuals with high arachnophobia levels. On the other hand, increasing the number of clusters could provide more nuanced results but may also introduce instability in the classification. In addition, the limited gender diversity and the lack of a sufficiently large and heterogeneous sample reduce the generalizability of the results and constrain the broader applicability of the model. These limitations are due to the nature of the available sample and the constraints of the current study, but they should be explicitly addressed in future research to enhance the robustness and relevance of the findings. The current findings also highlight the limitations of self-assessment tools such as the SPQ, which may not always provide an accurate measure of arachnophobia. Expert evaluations could complement self-assessments to provide a more objective view of each participant's phobia level. Further research could integrate these methods into other systems to personalize user experiences based on phobia levels. In games, new mechanics could be designed to dynamically adapt to the players' arachnophobia, creating evolving challenges that either accommodate or gradually expose them to phobic stimuli for a more immersive experience. Additionally, in medical contexts, VR-based BATs could provide a reliable tool for assessing and managing arachnophobia, leveraging VR's ability to simulate phobic stimuli without real-world risks.

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